

An AI-Enabled Digital Healthcare Workflow: Integrating Deep Learning and Large Language Models for Interpretable Wound Triage

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Abstract—Accurate chronic wound triage is critical in clinical settings, yet the adoption of deep learning in this domain is often limited by the opaque nature of AI models and the absence of deployable engineering workflows. This paper presents an end-to-end Artificial Intelligence-assisted Clinical Decision Support System (AI-CDSS) that integrates Convolutional Neural Networks (CNNs) with Large Language Models (LLMs) for interpretable wound triage. Operating solely on visual data to maximize point-of-care usability, the vision component employs a VGG16 backbone optimized through transfer learning and selective fine-tuning to classify images into six categories. The fine-tuned model achieved a global accuracy of 74.79%, with class-wise analysis showing that fine-tuning increased venous ulcer recall from 87.09% to 91.94% at the cost of a corresponding decline in pressure ulcer recall to 23.53%, a trade-off addressed through a permanent, always-visible safety disclosure rather than a dynamic confidence threshold. The classifier is embedded in a scalable, asynchronous microservices architecture. The system employs a generative LLM as a Narrative Interpretability layer, translating probabilistic outputs and patient metadata into structured clinical pre-reports that explicitly state model uncertainties to support clinician decision-making.

Index Terms—Digital Healthcare, Narrative Interpretability, Clinical Decision Support System, Chronic Wound Triage, Convolutional Neural Networks, Large Language Models.

I. INTRODUCTION

Accurate and timely assessment of chronic wounds, including diabetic foot, pressure, venous, and surgical ulcers, remains a persistent challenge in clinical practice due to their high prevalence and economic burden [1], [2]. Wound evaluation relies heavily on visual inspection, which is time-consuming and subject to inter-observer variability. Delays or misclassification may lead to suboptimal interventions and increased burden on patients and healthcare systems [3].

The expansion of telemedicine has increased demand for scalable digital workflows that support triage and documentation while preserving traceability and interpretability [4],

[5]. CNNs have achieved competitive performance in medical image classification and are increasingly explored for wound classification [6]–[8].

Moving from a standalone classifier to a clinically usable solution requires more than predictive performance. A digital healthcare system must support asynchronous processing, uncertainty communication, and clinician-centered outputs [9]. LLMs have been explored for natural-language summarization and report drafting in healthcare, but their use in decision support must be framed with explicit safeguards and separation from clinical judgment [10]–[12].

This work proposes an end-to-end AI-CDSS Proof of Concept (PoC) for digital health. To maximize point-of-care usability, a vision-only approach is adopted with a VGG16 backbone [13] trained via Transfer Learning and selective Fine-Tuning to classify wound images into six categories. The classifier is integrated into a modular, asynchronous microservices architecture (Figure 1) connected to a generative LLM [14] acting as a Narrative Interpretability layer, producing structured clinical pre-reports that explicitly state model uncertainties.

The primary contribution is the design, implementation, and evaluation of this platform, with methodological emphasis on a two-stage training strategy: Transfer Learning with pre-trained ImageNet weights, followed by targeted Fine-Tuning of deep convolutional blocks [13]. The class-wise trade-offs introduced by fine-tuning are empirically evaluated, and the end-to-end architecture is shown to compensate for algorithmic limitations through the generative interpretability layer.

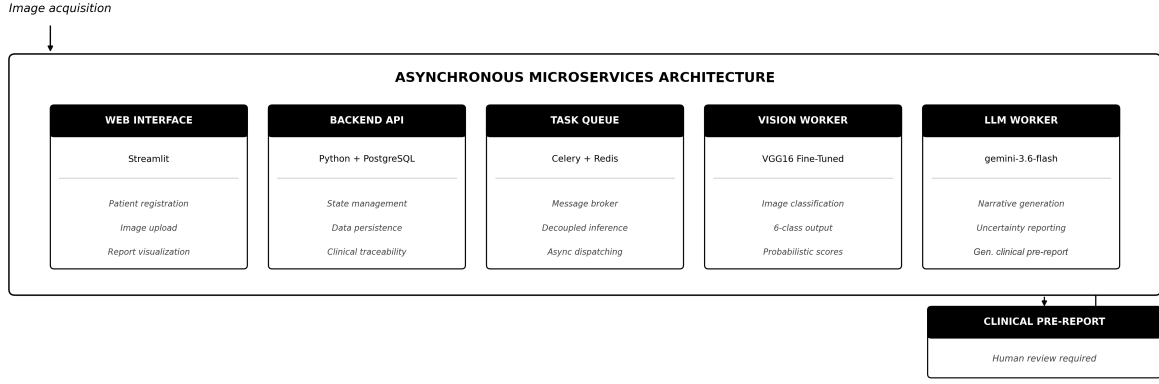


Fig. 1. Overview of the proposed AI-CDSS workflow, from visual data ingestion to LLM-generated pre-reports.

II. BACKGROUND AND RELATED WORK

A. CNNs for Ulcer Image Classification

CNNs have become the standard approach for medical image classification due to their ability to learn hierarchical visual representations from data [6]. In wound care, CNNs have been applied to classify chronic wounds from photographs captured under heterogeneous conditions typical of real-world clinical and telemedicine settings [7], [8], [15].

Anisuzzaman et al. [8] and Patel et al. [15] showed that fusing image data with body location coordinates improves classification accuracy for chronic wounds. Although effective, manually mapping spatial coordinates for each lesion introduces operational friction in fast-paced point-of-care workflows. This work diverges by adopting a vision-only approach, relying on targeted model optimization to compensate for the absence of spatial metadata. Per-class metrics including recall, precision, and F1-score are used to identify clinically relevant failure modes [16], and uncertainty communication is treated as a system design requirement [17].

B. Transfer Learning and Fine-Tuning

Transfer learning reduces data requirements by adapting representations learned from large-scale natural image datasets to the target medical domain [18], [19]. VGG16 remains a common baseline due to its straightforward architecture, facilitating reproducible comparisons in resource-constrained settings [13]. Fine-tuning introduces a trade-off between domain specialization and overfitting; class-wise analysis is recommended to assess whether performance gains are generalizable or reflect redistribution across classes [20], [21].

C. CDSS and LLMs in Health Applications

CDSS are grouped into knowledge-based and AI-based systems [5], [11]. In both cases, effective adoption depends on how outputs are communicated and how uncertainty is managed [9]. LLMs have been explored as a Narrative Interpretability layer downstream of predictive models, translating structured evidence into clinician-facing narratives that reduce

reporting burden while preserving traceability [12]. Concerns about hallucination and over-reliance motivate designs that constrain LLMs to reporting tasks with mandatory human-in-the-loop review [10], [22].

III. METHODOLOGY

A. Dataset and Preprocessing

This study used the public dataset “Multi-modal wound classification using images and locations” [8], [15], collected at the AZH Wound and Vascular Center. Spatial metadata was excluded to implement a vision-only workflow suitable for mobile point-of-care use [8]. The dataset is publicly available and anonymized, complying with ethical standards for secondary data analysis.

Images were captured with an iPad Pro and a Canon digital camera, presenting real-world variability in scale, illumination, and background. The six-class taxonomy and the exact per-class distribution are described in Table I. The dataset was partitioned into 696 training images and 234 test images following the original split [8]. The class distribution is imbalanced: the most represented class (V, 185 training images) outnumbers the least represented pathological class (P, 100 training images) by a factor of 1.85, which we revisit in Section IV as a contributing factor to the class-wise recall gap observed after fine-tuning. Data augmentation including rotations ($\pm 20^\circ$), width and height shifts (20%), shear distortions (20%), zoom variations (20%), and horizontal flipping was applied to mitigate overfitting and improve feature generalization [18], [20].

TABLE I
IMAGE CLASSES AND PER-CLASS SAMPLE COUNTS.

Class	Meaning	Train	Test
D	Diabetic Foot Ulcer	139	46
P	Pressure Ulcer	100	34
S	Surgical Wound	122	42
V	Venous Leg Ulcer	185	62
N	Normal Skin	75	25
BG	Background	75	25
Total		696	234

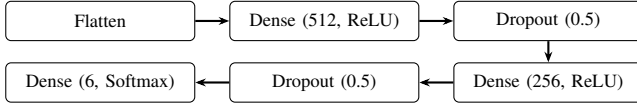


Fig. 2. Custom classification head appended to the VGG16 backbone after `block5_pool`.

B. Model Architecture and Two-Stage Training

VGG16 [13] was selected as the CNN backbone for its established performance and role as a reproducible baseline in medical imaging [6]. The original ImageNet classification head was replaced by a custom head appended after the last max-pooling layer (`block5_pool`), as illustrated in Figure 2.

Training proceeded in two stages. In the first stage (Transfer Learning), the convolutional base was frozen and the custom head was trained using the Adam optimizer with learning rate $lr = 1 \times 10^{-4}$, categorical cross-entropy loss, and Early Stopping with patience of 10 epochs. In the second stage (Fine-Tuning), the highest-level convolutional block (`block5`) was unfrozen and the network was retrained with a reduced learning rate ($lr = 1 \times 10^{-5}$) and stricter Early Stopping (patience of 5 epochs), allowing the filters to specialize in wound-specific visual textures while avoiding catastrophic forgetting of the pre-trained weights.

In both stages, Early Stopping and checkpoint selection monitored loss and accuracy computed on the 234-image test partition, since the original protocol did not carve out a separate internal validation split from the training data. We flag this explicitly as a methodological limitation: because the same partition used for model selection is also used for the final reported metrics, the selected checkpoint’s test accuracy is likely optimistic relative to a protocol with an independent validation split. A corrected protocol, holding out a stratified subset of the 696 training images for validation while leaving the test set untouched until final evaluation, is planned as a follow-up; we do not expect it to change the qualitative class-wise trends discussed in Section IV, though it may lower the reported point accuracy.

C. AI-CDSS System Architecture

The platform is orchestrated via Docker Compose across four containerized services sharing a common bridge network. The `backend` directory is not a standalone service; it is a shared Python module mounted as a volume into both the `streamlit` and `worker` containers, providing common access to the classification model, database operations, and LLM integration. A shared `bucket` volume provides persistent file storage for uploaded images and generated PDF reports. Table II describes each service.

The relational schema (Figure 3) maintains strict links between patient records, image assets, and AI-generated outputs. Every pre-report is therefore auditable and tied to the specific model checkpoint used during inference. Once the Celery worker computes the classification probabilities, the distribution is packaged with patient metadata and passed to

TABLE II
DOCKER COMPOSE SERVICES OF THE AI-CDSS ARCHITECTURE.

Service	Image / Stack	Role
<code>streamlit</code>	Streamlit	Web interface for patient registration, image upload, and report visualization.
<code>worker</code>	Celery	Asynchronous worker that loads the fine-tuned VGG16 model, executes image classification, invokes the LLM to produce the clinical pre-report.
<code>postgres</code>	PostgreSQL	Relational persistence of patient records, image metadata, classification results, and generated pre-reports.
<code>redis</code>	Redis	In-memory message broker connecting the web interface to the asynchronous worker.

a structured prompting pipeline. The generative component is Google’s `gemini-3.6-flash` [23], invoked through the `google-genai` SDK with default generation parameters (no explicit temperature or top-p override). The prompt template supplies the predicted class, the full probability vector, and the model’s historical precision, recall, and F1-score for that class, and instructs the LLM to produce a report of no more than 300 words. The LLM produces a clinical draft that states model uncertainties explicitly, presents secondary differential diagnoses, and is formatted as a preliminary summary explicitly labeled for clinician verification prior to any clinical use [10].

Because the deployed system persists real patient identifiers, clinical history, and images, unlike the pre-anonymized public dataset used for model training, it falls under Brazil’s General Data Protection Law (LGPD) [24]. The present proof of concept does not yet implement encryption at rest, authentication, or an anonymization pipeline for stored records; hardening the platform to LGPD requirements, including encrypted storage, role-based access control, audit logging, and a data-retention policy, is a prerequisite for any use beyond controlled evaluation and is prioritized as future engineering work.

The complete source code is publicly available.¹

¹<https://github.com/ErnaneJ/wound-ai-cdss>

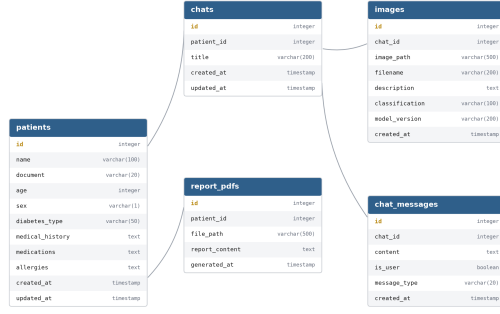


Fig. 3. Entity-relationship diagram of the AI-CDSS, illustrating the relational structure for clinical data persistence and auditability.

IV. RESULTS AND DISCUSSION

A. Incremental Model Performance

The vision engine was developed across three phases. Table III summarizes the performance gains at each stage.

Phase 1 (Epoch 50): Transfer Learning established a baseline at 72.22% accuracy. The learning curves in Figure 4 show stable convergence but a persistent gap between training and validation loss, indicating the classification head had not reached a stable plateau. The confusion matrix in Figure 5 shows high accuracy for non-pathological classes (BG and N), but a significant overlap between Pressure (P) and Venous (V) ulcers, suggesting the frozen convolutional base was insufficient to capture the fine-grained textural differences between these etiologies.

Phase 2 (Epoch 94): Extending training with a weight restoration protocol improved accuracy to 74.79%, as shown in Figure 6. The confusion matrix in Figure 7 reflects improved recall for Diabetic (D) and Surgical (S) classes, but class P recall remained at 29.41%, confirming that the P/V confusion is a structural limitation of the frozen feature extractor and justifying the subsequent fine-tuning stage.

Phase 3 (Fine-Tuning): Unfreezing block5 matched the extended transfer-learning accuracy (74.79%) while substantially redistributing recall across classes. The F1-score comparison in Figure 8 shows a substantial gain for class V, whose recall reached 91.94%. The final confusion matrix in Figure 9 confirms that venous ulcer triage reached high reliability. Class P recall decreased to 23.53%, a consequence of the model specializing in the dominant class distribution. Global accuracy alone therefore does not capture the effect of fine-tuning; the class-wise trade-off, gaining venous sensitivity at

TABLE III
PERFORMANCE GAINS ACROSS TRAINING STAGES.

Stage	Epoch	Accuracy	Recall (V)
Initial Baseline	50	72.22%	87.09%
Extended TL	94	74.79%	87.09%
Fine-Tuning	13 (+94)	74.79%	91.94%

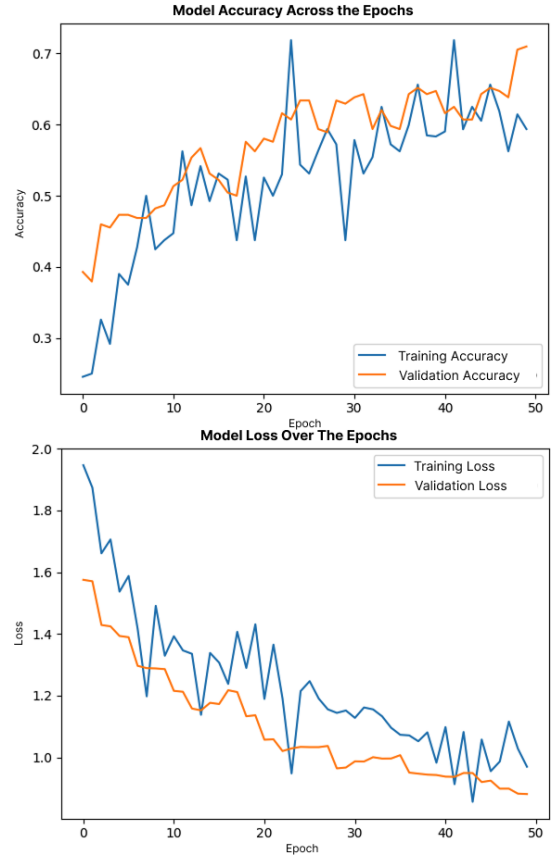


Fig. 4. Learning curves, Epochs 1–50. The gap between training and validation loss indicates room for further optimization.

the cost of pressure-ulcer sensitivity, is the actual outcome of this stage and is treated accordingly throughout this section. Notably, class P recall declines monotonically across all three phases (38.24% at Epoch 50, computed from Figure 5; 29.41% at Epoch 94; 23.53% after fine-tuning), indicating that the sensitivity loss reflects progressive specialization toward the dominant class distribution rather than an isolated effect of the fine-tuning stage.

We treat this reduction in class P recall as a patient-safety limitation, since pressure injuries are themselves a patient-safety indicator subject to prevention protocols and adverse-event reporting in Brazilian clinical services [3], not merely as an optimization trade-off. The current single-split protocol cannot reliably calibrate a per-prediction confidence threshold (Section IV-B), so the Narrative Interpretability layer instead embeds a static, always-visible disclosure of the model's documented class P sensitivity (recall of 23.53%) into every generated pre-report. Clinicians are alerted to this weakness regardless of the classifier's output for a given image, favoring a transparent, permanently visible safety disclosure over an uncalibrated dynamic trigger [17]. This disclosure complements, but does not substitute, validated structured risk-assessment instruments such as the Braden and Norton scales, and it does not replace institutional pressure-injury prevention protocols:

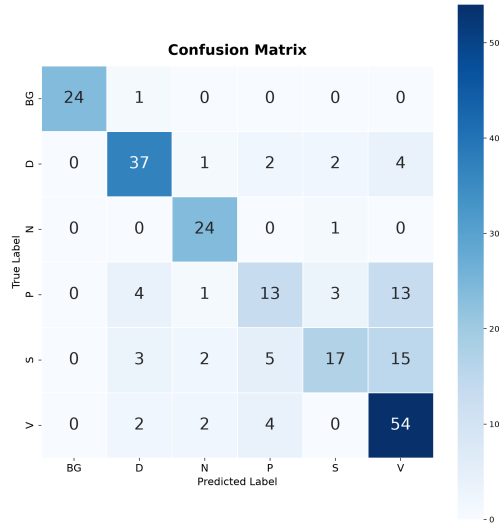


Fig. 5. Confusion matrix at Epoch 50. High accuracy for non-pathological classes, with notable overlap between P and V ulcers.

the classifier output is meant to prompt attention, while risk stratification and preventive care decisions remain governed by established clinical scales.

The global accuracy of 74.79% achieved by the fine-tuned model corresponds to a 95% Wilson confidence interval of [68.9%, 79.9%] ($n = 234$), reflecting the estimation uncertainty inherent to a test set of this size. This result is consistent with the literature under comparable constraints, though direct comparison is limited by differing preprocessing and data splits. Anisuzzaman et al. [8] reported 75.64% accuracy using whole images alone and 82.48% when fusing image and body-location data. Patel et al. [15], using a region-of-interest (ROI) cropping step and a 70/15/15 split, reported 85.41% accuracy without location data and 87.50% when adding location, a result driven in large part by ROI cropping rather than multi-modality alone. Relative to the whole-image, single-modality figure from Anisuzzaman et al., our result is comparable; the gap to the multimodal figures (7.7 to 12.7 percentage points) reflects both the cost of excluding spatial metadata and, in Patel et al.’s case, the absence of ROI cropping, which we did not adopt to preserve a simple, single-stage pipeline for point-of-care deployment. In this work, spatial metadata was intentionally excluded to keep the workflow viable at the point of care. The venous ulcer recall of 91.94% reflects the clinical priority assigned to that class, and the permanent, metrics-based safety disclosure for pressure ulcers addresses the sensitivity limitations that a global accuracy figure alone would not reveal.

B. Model Calibration

Clinical deployment requires calibrated confidence, not only point accuracy: predicted probabilities should reflect the true likelihood of correctness. We assessed calibration with the Expected Calibration Error (ECE) [25], using 10 equal-width confidence bins over an independent re-evaluation of the

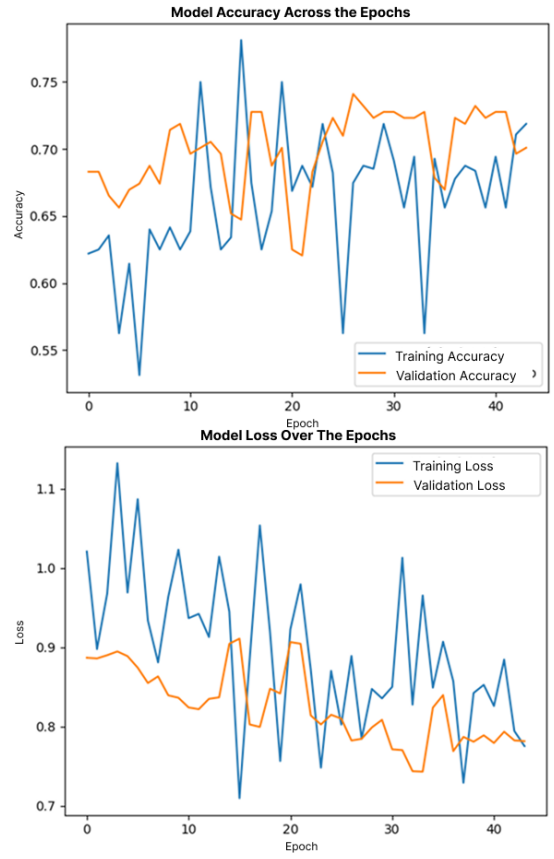


Fig. 6. Learning curves for the continuation phase (44 additional epochs, cumulative total of 94). Stable convergence achieved through automated weight restoration.

released model checkpoint on the 234-image test partition. This re-evaluation reproduced the 74.79% headline accuracy (175/234) reported in Table III, confirming that inference on the released checkpoint is deterministic given a fixed, non-shuffled test partition. Because Early Stopping and checkpoint selection are themselves single-run decisions (Section III), the reported point accuracy still carries the run-to-run variance of the training process, not of inference; repeating training across multiple random seeds and reporting mean $\pm$ standard deviation, rather than a single point estimate, is planned as future work.

The resulting ECE was 0.052, with a mean predicted confidence of 69.6% against an observed accuracy of 74.8%: the model is, on average, mildly underconfident rather than overconfident (Figure 10). Because the current protocol lacks a validation split independent from the test set (Section III), this ECE is reported as a diagnostic computed on the same data used for evaluation; we do not apply post-hoc temperature scaling here, since fitting a temperature parameter on the test set itself would reintroduce the same leakage. Recalibration on a proper held-out calibration split, obtained once the validation protocol described in Section III is corrected, is proposed as future work.



Fig. 7. Confusion matrix at Epoch 94 (74.79% accuracy). Class P recall remains at 29.41%.

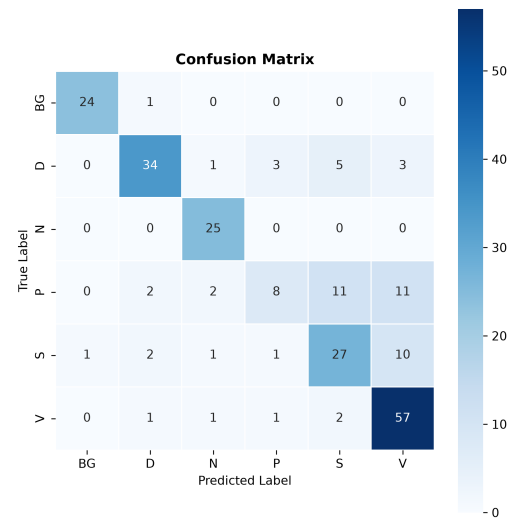


Fig. 9. Final fine-tuned confusion matrix. Venous (V) class recall reaches 91.94%.

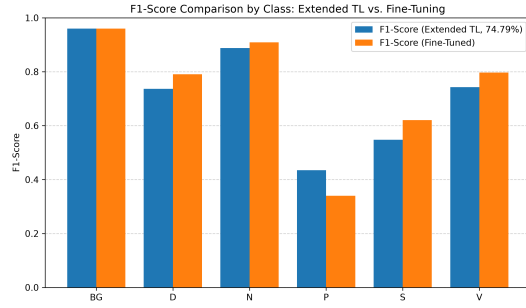


Fig. 8. Class-wise F1-score comparison: Extended Transfer Learning versus Fine-Tuned model. Both stages reach 74.79% global accuracy; the gain from fine-tuning is a redistribution of recall toward class V, not a change in the aggregate figure.

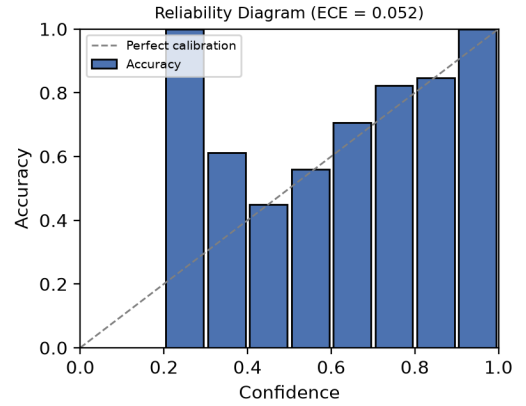


Fig. 10. Reliability diagram for the re-evaluated fine-tuned model (ECE = 0.052). Bars show observed accuracy per confidence bin; the dashed line marks perfect calibration.

C. Operational Validation

Figures 11 and 12 illustrate the operational interface using a synthetic patient record created solely for demonstration; no real patient data is shown. Figure 11 shows the patient registration interface, which allows structured clinical data entry alongside high-resolution image upload at the point of care.

Figure 12 shows the decision support interface. The system presents the CNN’s classification output translated by the LLM into a narrative clinical pre-report directly in the chat: the primary predicted class and confidence, a differential discussion of the leading secondary candidate classes drawn from the full probability vector, and the model’s documented class-wise limitations, including the reduced sensitivity for pressure ulcers, which are always appended to the report regardless of the current prediction. The AI is positioned as a documentation assistant, not a diagnostic authority; the pre-report is a preliminary, AI-generated draft intended for manual clinician review and is not a substitute for clinical judgment.

V. CONCLUSION

This paper presented a scalable and interpretable digital healthcare workflow for chronic wound triage, designed for the resource-constrained, point-of-care settings common in Brazilian public health services, where anatomical location metadata and specialized equipment are often unavailable. The proposed AI-CDSS integrates a VGG16 vision engine, an asynchronous microservices architecture, and a generative LLM interpretability layer. It delivers practical value under the constraints of a limited and imbalanced medical dataset and treats uncertainty communication as a first-class design requirement, not an afterthought.

The two-stage training strategy yielded a peak accuracy of 74.79% (95% CI [68.9%, 79.9%]) and a venous ulcer recall of 91.94% after fine-tuning, as detailed in the class-wise analysis. The analysis also exposed the trade-offs introduced by domain specialization: global accuracy metrics can

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